

Methods of solving a system of linear equations

Substitution and elimination — two exact methods that finish where the drawing could only estimate.

LESSON 147–149

3 × 40 MIN

MATEMATIKA 7, §63

BEYOND STAGE 8

LESSON CLOCK

10

30'

35'

30'

15'

Two exact methods	10 min
Substitution	30 min
Elimination	35 min
Multiplying first	30 min
No solution, or infinitely many	15 min

LEARNING OBJECTIVES

- Solve a system by substitution.
- Solve a system by adding or subtracting the equations.
- Multiply an equation through to make elimination possible.
- Recognise from the algebra when there is no solution or infinitely many.

TEXTBOOK

National	pp. 440–450
Cambridge	Stage 8 extension
Workbook	Extension sheet 2E

01

Explanation

Two exact methods

The graph showed *where* the answer is; these two methods produce it exactly. Both work by turning two equations in two unknowns into one equation in one unknown.

METHOD	THE MOVE	BEST WHEN
substitution	express one unknown and put it into the other equation	one unknown already stands alone
elimination	add or subtract the equations to cancel an unknown	a pair of coefficients match or are opposite

BOTH METHODS HAVE THE SAME AIM

Get to one equation with one letter in it. Everything already known about linear equations then finishes the job, and the second unknown comes from back-substitution.

Substitution

STEP	WORKING FOR $\{ Y = X - 1 ; X + Y = 5 \}$
1 — express one unknown	$y = x - 1$
2 — substitute into the other	$x + (x - 1) = 5$
3 — solve	$2x = 6, x = 3$
4 — back-substitute	$y = 3 - 1 = 2$
5 — check both	$3 + 2 = 5 \checkmark, 3 - 2 = 1 \checkmark$

SUBSTITUTE INTO THE OTHER EQUATION

Putting $y = x - 1$ back into $y = x - 1$ gives $0 = 0$ and no information at all. The expression must go into the equation it did not come from.

Elimination

When the coefficients of one unknown are equal, subtract the equations; when they are opposite, add them.

SYSTEM	MOVE	RESULT	SOLUTION
$x + y = 5, x - y = 1$	add	$2x = 6$	$(3; 2)$
$2x + y = 7, x + y = 4$	subtract	$x = 3$	$(3; 1)$
$3x + 2y = 12, 3x - 2y = 0$	add	$6x = 12$	$(2; 3)$

SUBTRACTING MEANS SUBTRACTING EVERY TERM

From $2x + y = 7$ take $x + y = 4$: the right-hand sides give $7 - 4 = 3$, not 7. Line the two equations up and work column by column.

Multiplying first

When no coefficients match, multiply one equation — or both — through by a number until a pair of them does. Multiplying every term of an equation by the same non-zero number leaves its solutions untouched.

STEP	WORKING FOR $\{ 2X + 3Y = 12 ; X - Y = 1 \}$
multiply the second by 3	$3x - 3y = 3$
now the y coefficients are opposite	$+3y$
add the two equations	$5x = 15$
solve and back-substitute	$x = 3, y = 2$
check both	$6 + 6 = 12 \checkmark, 3 - 2 = 1 \checkmark$

Sometimes both equations need multiplying:

STEP	WORKING FOR $\{ 2X + 5Y = 16 ; 3X - 2Y = 5 \}$
multiply the first by 2	$4x + 10y = 32$
multiply the second by 5	$15x - 10y = 25$
add	$19x = 57$
solve and back-substitute	$x = 3, y = 2$

MULTIPLY EVERY TERM, BOTH SIDES INCLUDED

Turning $x - y = 1$ into $3x - 3y = 1$ changes the equation and loses the solution. The right-hand side is multiplied too.

No solution, or infinitely many

SYSTEM	THE ALGEBRA GIVES	MEANING
$y = 2x, y = 2x + 3$	$0 = 3$	no solution — parallel lines
$y = 2x, 2y = 4x$	$0 = 0$	infinitely many — the same line
$x + y = 5, x - y = 1$	$x = 3$	one solution

THE ALGEBRA REPORTS THE GEOMETRY

Reaching a false statement means the lines never meet; reaching an always-true one means they are the same line. Neither is a mistake — each is the answer, and it should be written in words.

02 Worked examples

EXAMPLE 1 Solve $\{y = x - 1 ; x + y = 5\}$ by substitution.

① y is already expressed.

② $x + (x - 1) = 5$
Into the other equation.

③ $2x = 6$, so $x = 3$.

④ $y = 2$. Check: $3 + 2 = 5$ ✓

ANSWER

$(3; 2)$

EXAMPLE 2 Solve $\{2x + y = 7 ; x + y = 4\}$ by elimination.

① The coefficients of y are equal.

② Subtract: $(2x - x) + (y - y) = 7 - 4$.

③ $x = 3$

④ $3 + y = 4$, so $y = 1$.

ANSWER

$(3; 1)$

EXAMPLE 3 Solve $\{ 2x + 3y = 12 ; x - y = 1 \}$.

- ① From the second, $x = y + 1$.
A coefficient of 1 invites substitution.

② $2(y + 1) + 3y = 12$

③ $5y = 10$, so $y = 2$.

④ $x = 3$. Check: $6 + 6 = 12$ ✓

ANSWER

$(3; 2)$

03 Interactive model

Solve one system by both methods side by side on the board; the class chooses its own favourite and, more usefully, learns to see which system suits which.

Which method, and what does it give?

$\{ y = x - 1 ; x + y = 5 \}$ suits:

substitution

elimination

drawing only

neither

$\{ x + y = 5 ; x - y = 1 \}$ suits:

substitution

elimination by adding

elimination by subtracting

drawing only

$\{ 2x + y = 7 ; x + y = 4 \}$ suits:

adding

subtracting

multiplying first

drawing

Adding $x + y = 5$ and $x - y = 1$ gives:

$$2x = 6$$

$$2y = 4$$

$$2x = 4$$

$$x = 6$$

Then x is:

2

3

4

6

And y is:

1

2

3

5

Reaching $0 = 3$ means:

one solution

no solution

infinitely many

an error

Reaching $0 = 0$ means:

one solution

no solution

infinitely many

an error

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Substitution method	O'rniga qo'yish usuli	Способ подстановки
Elimination method	Qo'shish usuli	Способ сложения
To express	Ifodalash	Выразить
To eliminate	Yo'qotmoq	Исключить
Opposite coefficients	Qarama-qarshi koeffitsiyentlar	Противоположные коэффициенты
To multiply through	Ko'paytirib chiqish	Умножить обе части
Back-substitution	Teskari o'rniga qo'yish	Обратная подстановка
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Substitution puts an expression into:

the same equation

the other equation

both

neither

Equal coefficients call for:

adding

subtracting

multiplying

drawing

Opposite coefficients call for:

adding

subtracting

dividing

nothing

$\{ 2x + y = 7 ; x + y = 4 \}$ has solution:

(3; 1)

(1; 3)

(2; 2)

(4; 0)

After finding one unknown you should:

stop

back-substitute

draw the graph

multiply

The final check uses:

one equation

both equations

the graph

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\{ y = x - 1 ; x + y = 5 \}$

ANSWER $(3; 2)$

2. $\{ x + y = 5 ; x - y = 1 \}$

ANSWER $(3; 2)$

3. $\{ 2x + y = 7 ; x + y = 4 \}$

ANSWER $(3; 1)$

4. $\{ y = 2x ; x + y = 6 \}$

ANSWER $(2; 4)$

5. $\{ x = 3 ; x + y = 8 \}$

ANSWER $(3; 5)$

6. $\{ y = 5 ; 2x + y = 11 \}$

ANSWER $(3; 5)$

7. Adding $x + y = 5$ and $x - y = 1$

ANSWER $2x = 6$

Medium · the standard question

7 PROBLEMS

1. $\{ 2x + 3y = 12 ; x - y = 1 \}$

ANSWER (3; 2)

2. $\{ 3x + 2y = 12 ; 3x - 2y = 0 \}$

ANSWER (2; 3)

3. $\{ x + 2y = 8 ; 3x - 2y = 0 \}$

ANSWER (2; 3)

4. $\{ 2x - y = 3 ; x + y = 3 \}$

ANSWER (2; 1)

5. $\{ y = 2x ; y = 2x + 3 \}$

ANSWER No solution

6. $\{ y = 2x ; 2y = 4x \}$

ANSWER Infinitely many

7. $\{ 4x + y = 14 ; x - y = 1 \}$

ANSWER (3; 2)

Hard · stretch and reason

7 PROBLEMS

1. $\{ 3x + 2y = 16 ; 5x - 2y = 8 \}$

ANSWER (3; 3.5)

2. $\{ 2x + 5y = 16 ; 3x - 2y = 5 \}$

ANSWER (3; 2)

3. $\{ 5x - 3y = 9 ; 2x + 3y = 12 \}$

ANSWER (3; 2)

4. For which k has $\{ y = kx + 1 ; y = 3x + 5 \}$ no solution?

ANSWER $k = 3$

5. $\{ \frac{x}{2} + y = 4 ; x - y = 2 \}$

ANSWER (4; 2)

6. Why does substituting an expression into its own equation fail?

ANSWER It gives $0 = 0$ and no information

7. $\{ x + y = 10 ; x - y = 10 \}$

ANSWER (10; 0)

07 Homework

Homework — 5 tasks

Say which method you chose and why, then check the pair in both equations.

1. Solve $\{ y = x + 2 ; 2x + y = 11 \}$ by substitution.
2. Solve $\{ x + y = 9 ; x - y = 3 \}$ by elimination.
3. Solve $\{ 3x + y = 11 ; x + y = 5 \}$.
4. Solve $\{ 2x + 3y = 13 ; x - y = -1 \}$.

5. Decide how many solutions $\{ y = 4x - 1 ; y = 4x + 2 \}$ has, and explain.